

Supplementary Material

Household Bubbling Strategies for Epidemic Control and Social Connectivity

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I Demographic data

In this section, we briefly summarize the demographic data used to construct the household-based networks for the countries analyzed in this work. As described in the main text, the household size distribution $P(s)$ and the probability $P(w|s)$ of finding w workers in a household of size s are obtained from IPUMS International microdata. Here, we present these distributions for Argentina, China, Israel, and Spain, which are later used to generate the corresponding household networks.

Figure Ia shows the household size distribution $P(s)$ for the four countries. Figure Ib reports the average number of workers $\langle w(s) \rangle = \sum_w wP(w|s)$ as a function of household size s . Finally, in Fig. IIa-d, we display the distribution of workers within households $P(w|s)$.

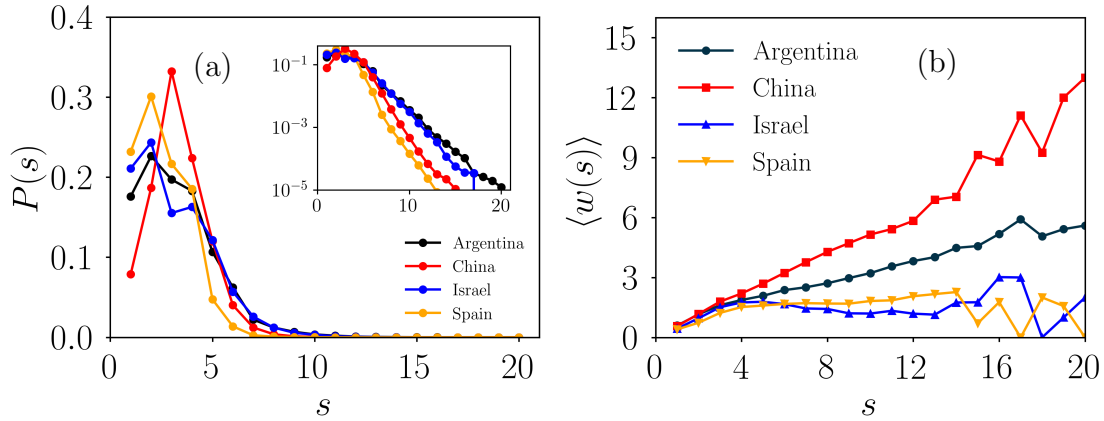


Figure I: Panel a: Household size distribution $P(s)$ for Argentina (black), China (red), Israel (blue), and Spain (orange). In the inset, we display the same curves but on a log-linear scale. Panel b: Average number of workers $\langle w(s) \rangle$ in a household of size s for Argentina (black), China (red), Israel (blue), and Spain (orange).

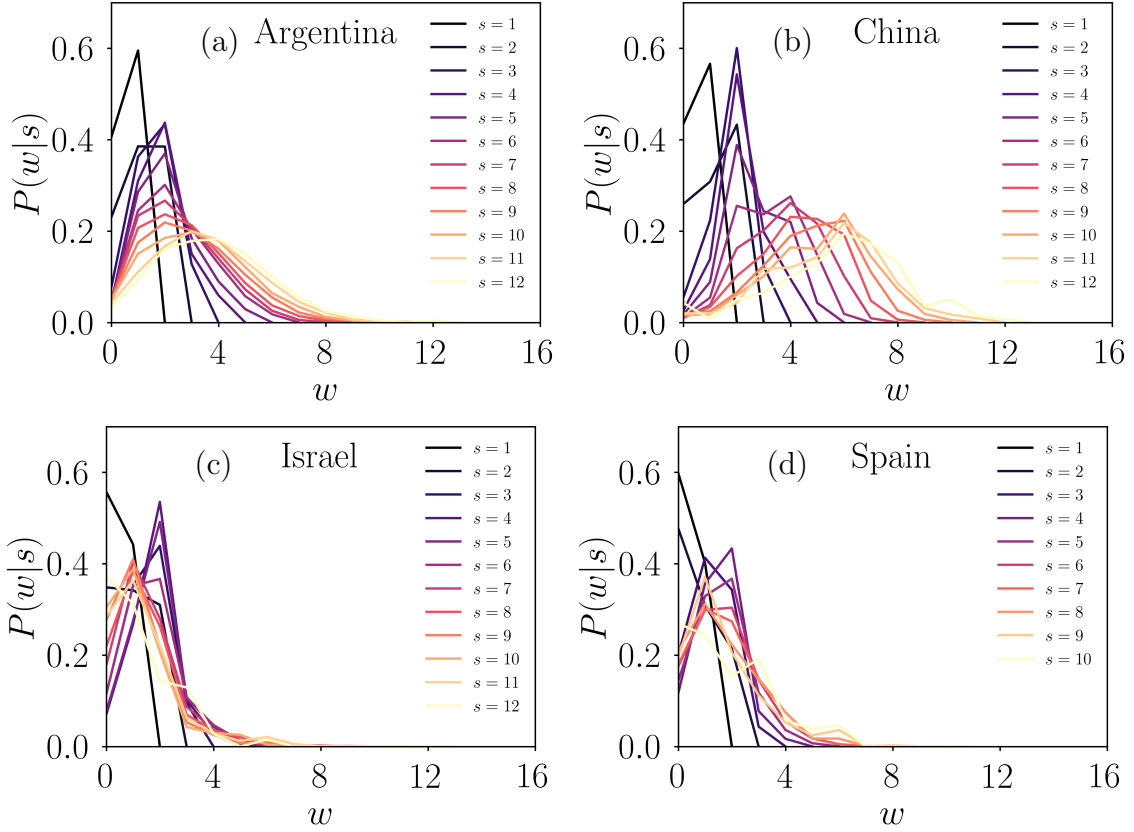


Figure II: $P(w|s)$ for Argentina (panel a), China (panel b), Israel (panel c), and Spain (panel d).

II Results for Argentina

In this section, we present additional results for Argentina. We first complete the analysis for the truncated power-law distribution $P(k_E) = PL(2.5, 1, 100)$ by considering the different household merging strategies introduced in the main text. We then present additional results for other external connectivity distributions to show that the theoretical predictions remain in good agreement with the simulation results.

II.1 Results for $PL(2.5, 1, 100)$

In this subsection, we consider a truncated power-law distribution for the number of external contacts, $P(k_E) = PL(2.5, 1, 100)$.

Figure III summarizes the simulation results obtained under the different household merging strategies described in the main text (see Sec. "Merging strategy").

For a more quantitative comparison across merging scenarios, Table 1 reports the critical external transmission probability β_c^E (computed for $\beta^I = 1$), together with the corresponding average number of internal links per household $\langle \ell_I \rangle$ and the fraction of the population whose bubble size increases as a result of the merging process. The case without merging is included as a reference.

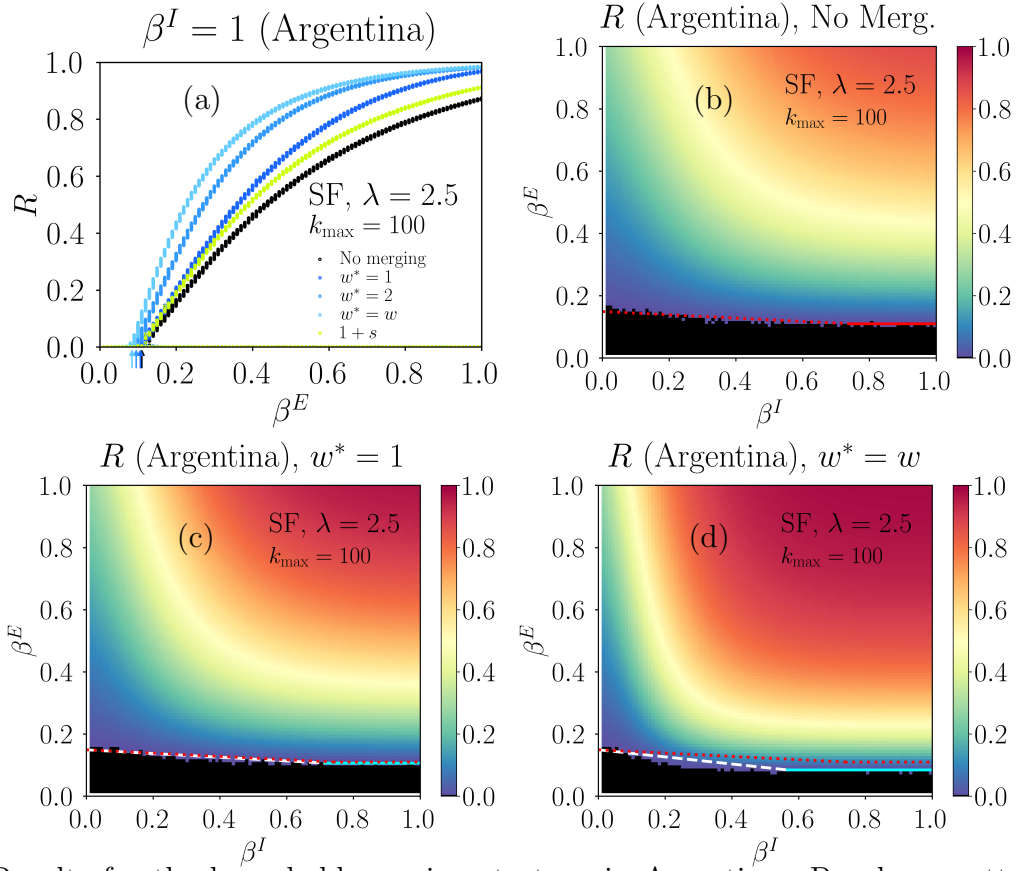


Figure III: Results for the household merging strategy in Argentina. Panel a: scatter-plot of the fraction of recovered individuals R at the final stage as a function of β^E (with $\beta^I = 1$) for several merging strategies, including the case without merging. Results are obtained from 1000 realizations on networks with $N = 5 \times 10^5$ individuals. Panels b-d: Heatmaps of the final fraction of recovered individuals in the $\beta^I - \beta^E$ plane, averaged over 400 realizations (excluding realizations where $R < 0.5\%$). Panel b corresponds to the case without merging. The solid red line corresponds to the theoretical prediction of the critical line presented in Eq. (2) in the main text, while the dotted red line shows the prediction from Eq. (4). Panel c: Scenario $w^* = 1$. Panel d: Scenario $w^* = w$. In panels c and d, the solid light blue lines represent the theoretical predictions from Eq. (2) for the merging scenarios, while the white dotted lines correspond to predictions from Eq. (4) with merging. The dotted red lines in panels c and d reproduce the theoretical curves from panel b (no merging case) for comparison.

Table 1: Summary of the main results for the different household merging strategies in **Argentina**. All reported values correspond to the case $P(k_E) = PL(2.5, 1, 100)$. For each scenario, we report the critical external transmission probability β_c^E (for $\beta^I = 1$), the average number of internal links $\langle \ell_I \rangle$, and the fraction of the population whose bubble size increases after merging. The case without merging is included for comparison.

Scenario	β_c^E	$\langle \ell_I \rangle$	f_{grow}
No merging	0.110	5.4	—
$w^* = 1$	0.106	9.7	41%
$w^* = 2$	0.095	15.8	76%
$w^* = w$	0.084	21.4	100%
$1 + s$	0.105	7.3	25%
$2 + s$	0.093	13.7	73%

II.2 Results for other external connectivity distributions

To further test the theoretical predictions presented in the main text, we also performed simulations using the following external connectivity distributions:

- $Pois(3, 1, 20)$, where $\langle k_E \rangle = 3.16$, $\langle k_E^2 \rangle = 12.6$, and $\chi_E = 0.35$,
- $Pois(4, 1, 20)$, where $\langle k_E \rangle = 4.07$, $\langle k_E^2 \rangle = 20.4$, and $\chi_E = 0.25$.
- $PL(2.5, 1, 20)$, where $\langle k_E \rangle = 1.63$, $\langle k_E^2 \rangle = 5.69$, and $\chi_E = 0.26$
- $PL(3, 1, 100)$, where $\langle k_E \rangle = 1.36$, $\langle k_E^2 \rangle = 4.32$, and $\chi_E = 0.29$.

Figure IV shows the heatmaps for the case without household merging. As in the previous results, the theoretical critical curves are in good agreement with the transition observed in the simulations for all the distributions considered.

Table 2 summarizes the values of the critical external transmission probability β_c^E (computed for $\beta^I = 1$) for all the merging strategies. For every external connectivity distribution considered, the $w^* = 1$ strategy gives the critical value closest to the case without merging.

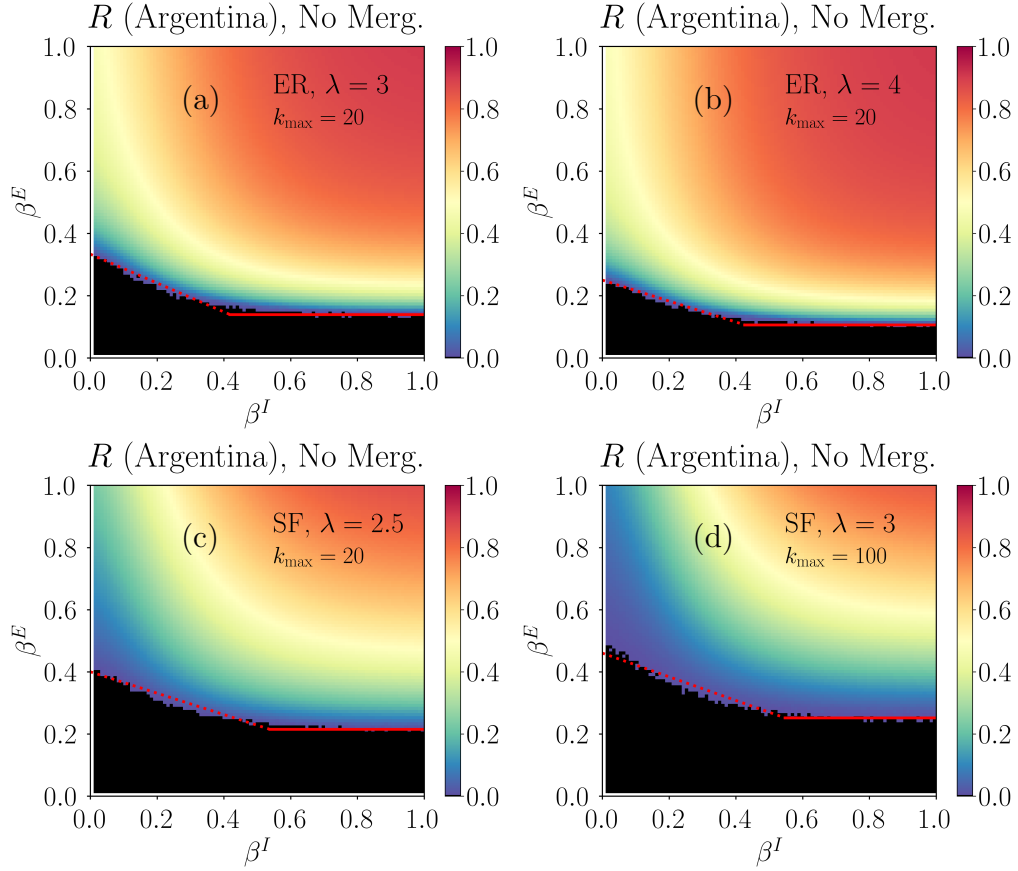


Figure IV: Heatmaps of the final fraction of recovered individuals in the $\beta^I - \beta^E$ plane for different external connectivity distributions without household merging. Panel a: $Pois(3, 1, 20)$, averaged over 1000 realizations on networks with $N = 5 \times 10^5$ individuals. Panel b: $Pois(4, 1, 20)$, averaged over 1000 realizations on networks with $N = 5 \times 10^5$ individuals. Panel c: $PL(2.5, 1, 20)$, averaged over 400 realizations on networks with $N = 2 \times 10^6$ individuals. Panel d: $PL(3, 1, 100)$, averaged over 900 realizations on networks with $N = 2 \times 10^6$ individuals.

Table 2: Critical external transmission probability β_c^E (for $\beta^I = 1$) obtained for different external connectivity distributions and household merging strategies.

Scenario	$Pois(3, 1, 20)$	$Pois(4, 1, 20)$	$PL(2.5, 1, 20)$	$PL(3, 1, 100)$
No merging	0.139	0.107	0.215	0.252
$w^* = 1$	0.130	0.099	0.203	0.238
$w^* = 2$	0.103	0.079	0.168	0.198
$w^* = w$	0.083	0.064	0.140	0.165
$1 + s$	0.126	0.097	0.199	0.233
$2 + s$	0.099	0.076	0.162	0.191

III Results for China

In this section, we present supplementary results for China, considering both homogeneous and heterogeneous external degree distributions $P(k_E)$. External connections are modeled using either a truncated Poisson distribution, $Pois(2, 1, 5)$ (homogeneous case), or a truncated power-law distribution, $PL(2.5, 1, 100)$ (heterogeneous case).

Figure V reports the results obtained for the homogeneous case. Panel (a) shows the final fraction of recovered individuals, as a function of the external infection probability (for $\beta^I = 1$), and for several household merging strategies, including the case without merging. Panels (b–d) display heatmaps of R in the $\beta^I - \beta^E$ plane for several merging strategies.

The corresponding results for the heterogeneous external degree distribution are shown in Fig. VIa–d.

A quantitative comparison of the different household merging strategies is provided in Table 3. For each scenario, we report the critical external infection probability for both homogeneous and heterogeneous external connections (computed for $\beta^I = 1$), together with the average number of internal links per household $\langle \ell_I \rangle$ and the fraction of the population whose bubble size increases after merging. The case without merging is included as a reference.

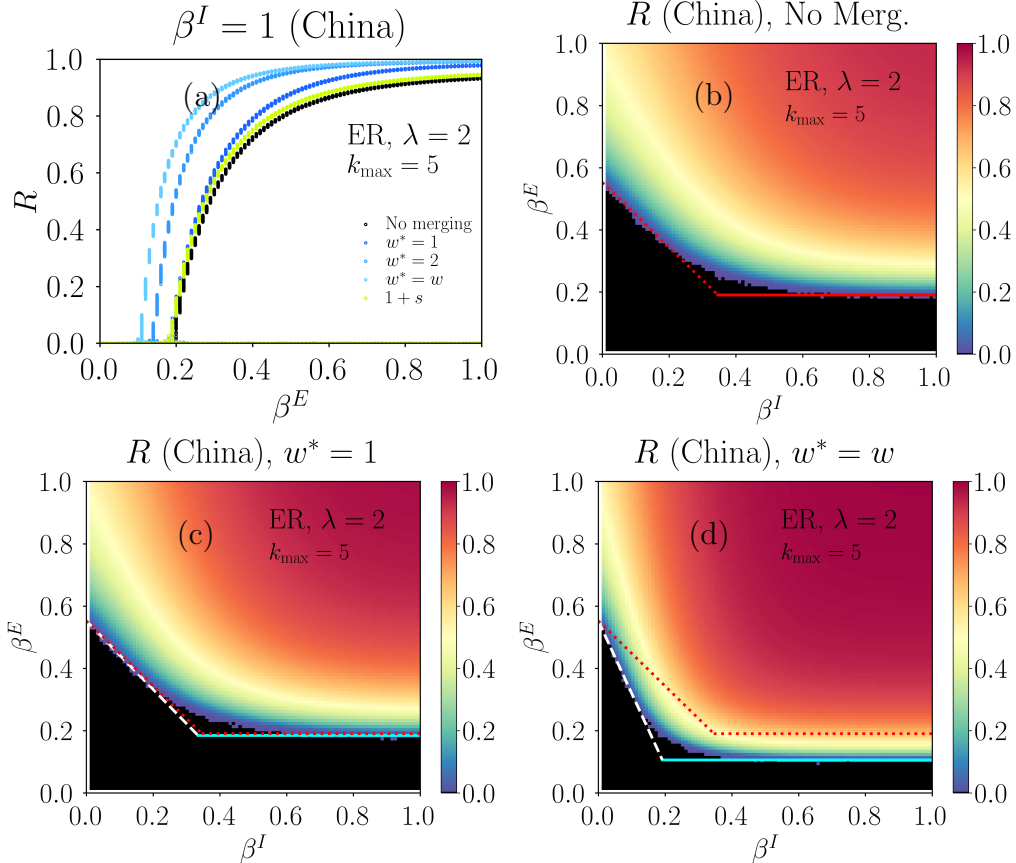


Figure V: Results for the household merging strategy in China. Panel a: scatter-plot of the fraction of recovered individuals R at the final stage as a function of β^E (with $\beta^I = 1$) for several merging strategies, including the case without merging. Panels b–d: Heatmaps of the final fraction of recovered individuals in the $\beta^I - \beta^E$ plane, for the following cases: No merging (panel b), $w^* = 1$ (panel c), and $w^* = w$ (panel d).

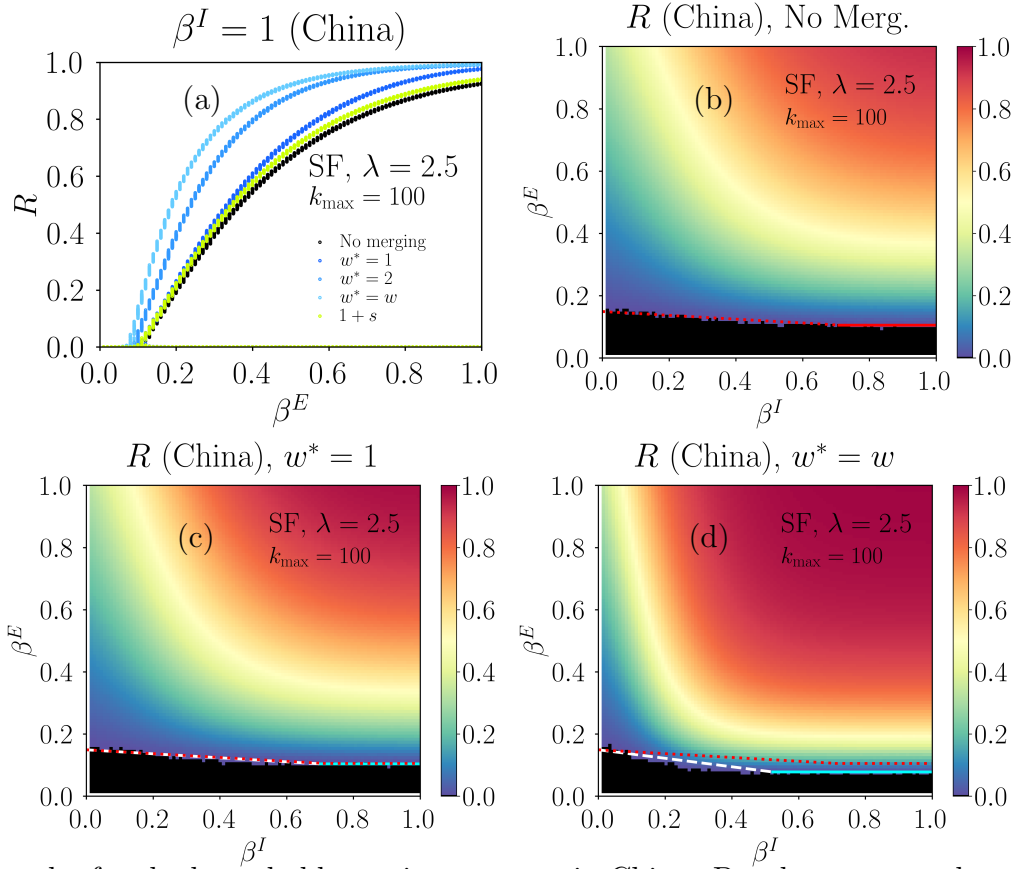


Figure VI: Results for the household merging strategy in China. Panel a: scatter-plot of the fraction of recovered individuals R at the final stage as a function of β^E (with $\beta^I = 1$) for several merging strategies, including the case without merging. Panels b-d: Heatmaps of the final fraction of recovered individuals in the $\beta^I - \beta^E$ plane, for the following cases: No merging (panel b), $w^* = 1$ (panel c), and $w^* = w$ (panel d).

Table 3: Summary of the main results for the different household merging strategies in **China**. For each scenario, we report: 1) the critical external transmission probability β_c^E (for $\beta^I = 1$) for homogeneous external connections (ER), and heterogeneous (SF), 2) the average number of internal links $\langle \ell_I \rangle$, and 3) the fraction of the population whose bubble size increases after merging. The case without merging is included for comparison.

Scenario	β_c^E ($Pois(2, 1, 5)$)	β_c^E ($PL(2.5, 1, 100)$)	$\langle \ell_I \rangle$	f_{grow}
No merging	0.191	0.106	4.7	—
$w^* = 1$	0.184	0.104	6.9	24%
$w^* = 2$	0.139	0.091	13.7	71%
$w^* = w$	0.106	0.078	20.6	100%
$1 + s$	0.184	0.104	5.5	10%
$2 + s$	0.151	0.095	8.9	44%

IV Results for Israel

In this section, we present supplementary results for Israel, considering both homogeneous and heterogeneous external degree distributions $P(k_E)$.

Figure VII reports the results obtained for the homogeneous case. Panel (a) shows the final fraction of recovered individuals, as a function of the external infection probability (for $\beta^I = 1$), and for several household merging strategies, including the case without merging. Panels (b–d) display heatmaps of R in the $\beta^I - \beta^E$ plane for several scenarios.

The corresponding results for the heterogeneous external degree distribution are shown in Fig. VIIIa–d.

Finally, a quantitative comparison of the different household merging strategies is provided in Table 4.

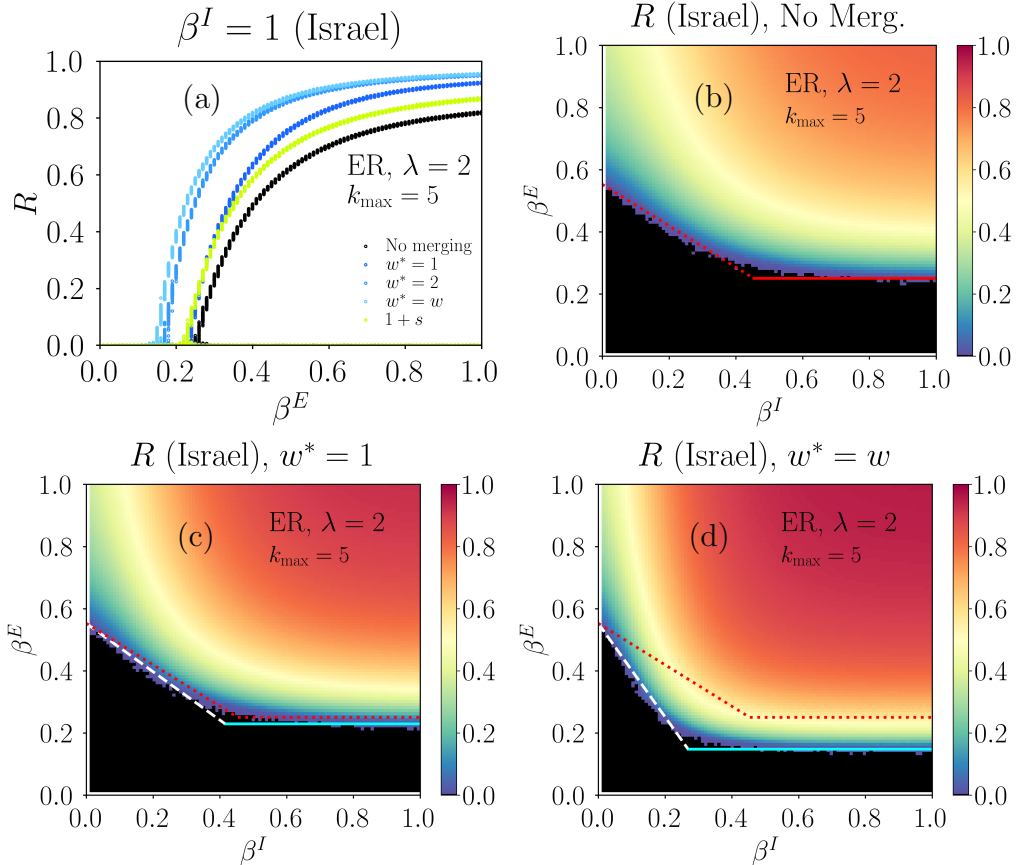


Figure VII: Results for the household merging strategy in Israel. Panel a: scatter-plot of the fraction of recovered individuals R at the final stage as a function of β^E (with $\beta^I = 1$) for several merging strategies, including the case without merging. Panels b–d: Heatmaps of the final fraction of recovered individuals in the $\beta^I - \beta^E$ plane, for the following cases: No merging (panel b), $w^* = 1$ (panel c), and $w^* = w$ (panel d).

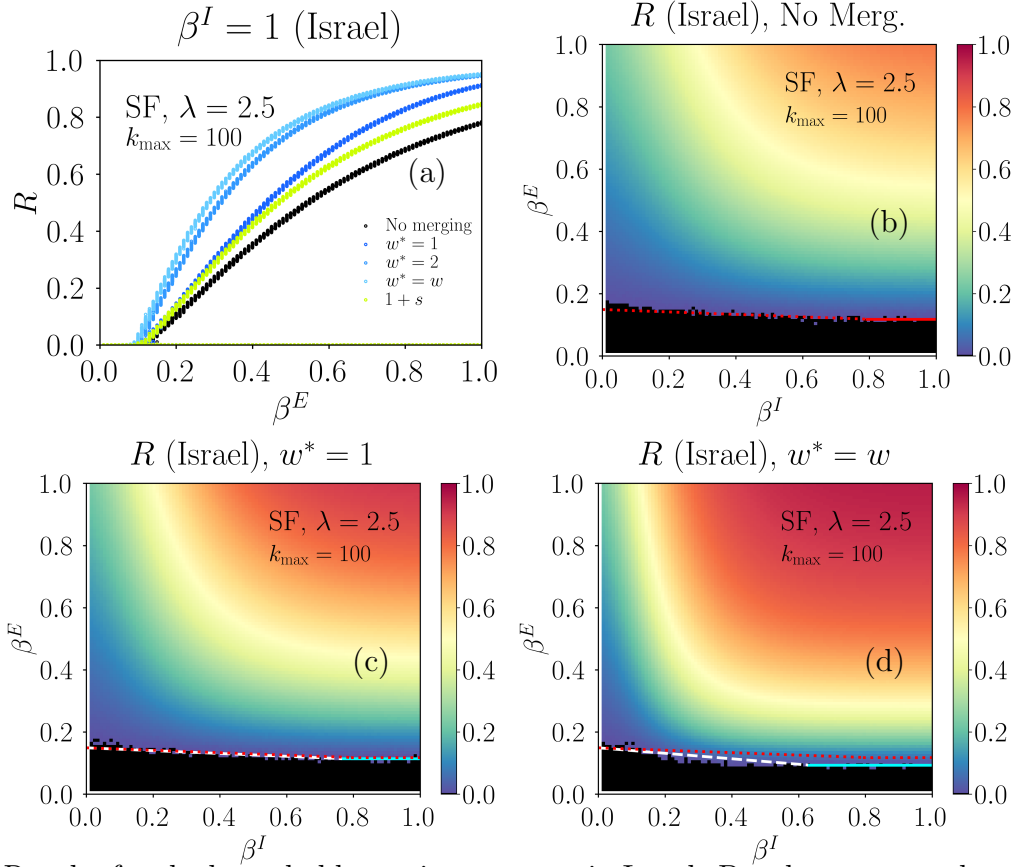


Figure VIII: Results for the household merging strategy in Israel. Panel a: scatter-plot of the fraction of recovered individuals R at the final stage as a function of β^E (with $\beta^I = 1$) for several merging strategies, including the case without merging. Panels b-d: Heatmaps of the final fraction of recovered individuals in the $\beta^I - \beta^E$ plane, for the following cases: No merging (panel b), $w^* = 1$ (panel c), and $w^* = w$ (panel d).

Table 4: Summary of the main results for the different household merging strategies in **Israel**. For each scenario, we report: 1) the critical external transmission probability β_c^E (for $\beta^I = 1$) for homogeneous (ER) and heterogeneous (SF) external connections, 2) the average number of internal links $\langle \ell_I \rangle$, and 3) the fraction of the population whose bubble size increases after merging. The case without merging is included for comparison.

Scenario	β_c^E ($Pois(2, 1, 5)$)	β_c^E ($PL(2.5, 1, 100)$)	$\langle \ell_I \rangle$	f_{grow}
No merging	0.250	0.118	5.2	—
$w^* = 1$	0.230	0.114	10.4	51%
$w^* = 2$	0.169	0.100	17.6	89%
$w^* = w$	0.149	0.094	20.2	100%
$1 + s$	0.223	0.113	7.5	32%
$2 + s$	0.168	0.100	15.2	87%

V Results for Spain

In this section, we present supplementary results for Spain, considering both homogeneous and heterogeneous external degree distributions $P(k_E)$.

Figure IX reports the results obtained for the homogeneous case. Panel (a) shows the final fraction of recovered individuals, as a function of the external infection probability (for $\beta^I = 1$), and for several household merging strategies, including the case without merging. Panels (b–d) display heatmaps of R in the $\beta^I - \beta^E$ plane for several scenarios.

The corresponding results for the heterogeneous external degree distribution are shown in Fig. Xa–d.

Finally, a quantitative comparison of the different household merging strategies is provided in Table 5. We did not include the strategy $2 + s$ because the fraction of households of size $s \leq 2$ is greater than the fraction of households of size $s > 2$.

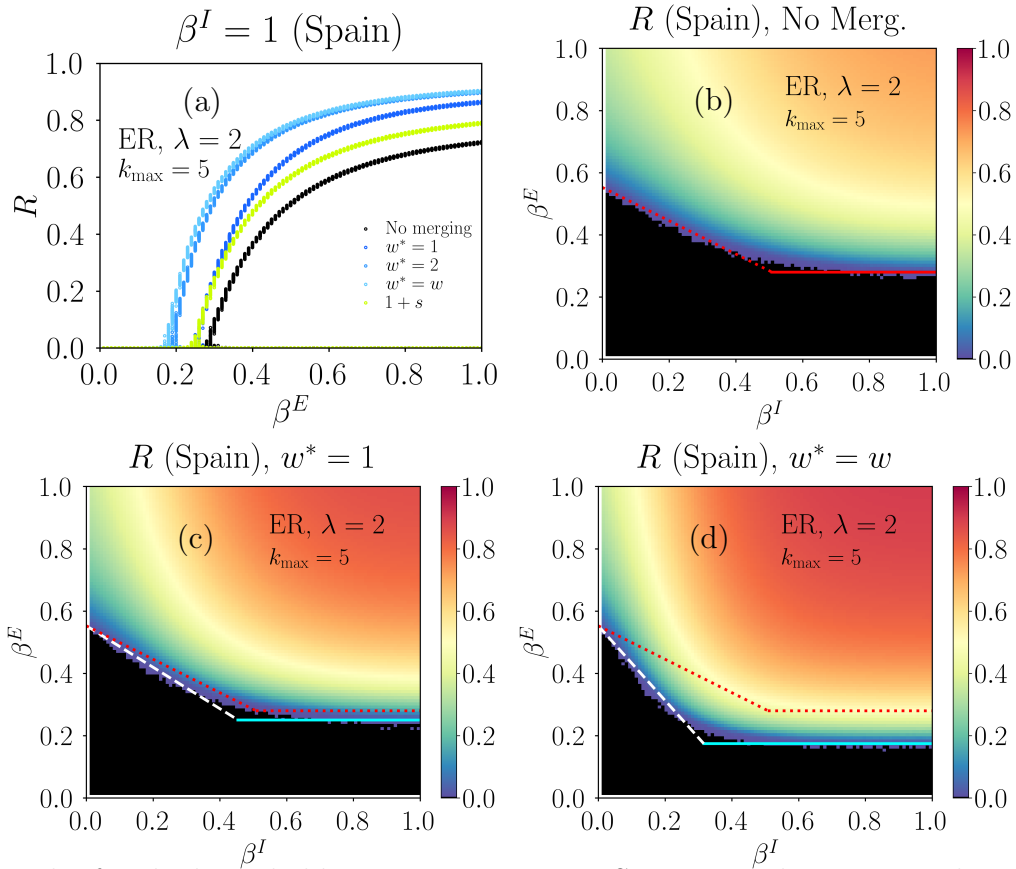


Figure IX: Results for the household merging strategy in Spain. Panel a: scatter-plot of the fraction of recovered individuals R at the final stage as a function of β^E (with $\beta^I = 1$) for several merging strategies, including the case without merging. Panels b–d: Heatmaps of the final fraction of recovered individuals in the $\beta^I - \beta^E$ plane, for the following cases: No merging (panel b), $w^* = 1$ (panel c), and $w^* = w$ (panel d).

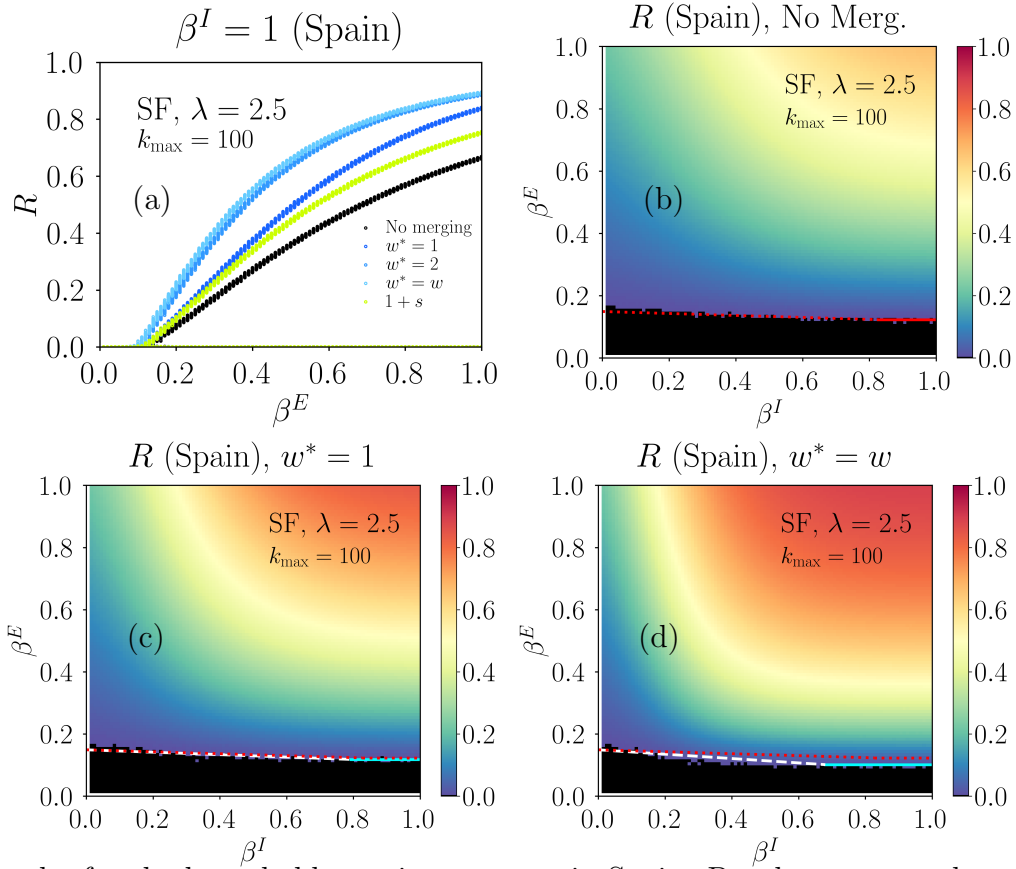


Figure X: Results for the household merging strategy in Spain. Panel a: scatter-plot of the fraction of recovered individuals R at the final stage as a function of β^E (with $\beta^I = 1$) for all merging strategies, including the case without merging. Panels b-d: Heatmaps of the final fraction of recovered individuals in the $\beta^I - \beta^E$ plane, for the following cases: No merging (panel b), $w^* = 1$ (panel c), and $w^* = w$ (panel d).

Table 5: Summary of the main results for the different household merging strategies in **Spain**. For each scenario, we report: 1) the critical external transmission probability β_c^E (for $\beta^I = 1$) for homogeneous (ER) and heterogeneous (SF) external connections, 2) the average number of internal links $\langle \ell_I \rangle$, and 3) the fraction of the population whose bubble size increases after merging. The case without merging is included for comparison.

Scenario	β_c^E ($Pois(2, 1, 5)$)	β_c^E ($PL(2.5, 1, 100)$)	$\langle \ell_I \rangle$	f_{grow}
No merging	0.280	0.123	2.8	—
$w^* = 1$	0.251	0.118	7.3	63%
$w^* = 2$	0.189	0.105	11.3	94%
$w^* = w$	0.174	0.101	12.3	100%
$1 + s$	0.247	0.117	4.6	36%